LECTURE 5

Proposition — <u>Geometric Series Test</u>

For $\sum_{n=l}^{\infty} ar^n$ if $|r| \geq 1$, it <u>diverges</u>.

If $|r| < 1$, it <u>converges</u> to $\frac{ar^l}{1-r}$

Proposition — <u>Integral Test</u>

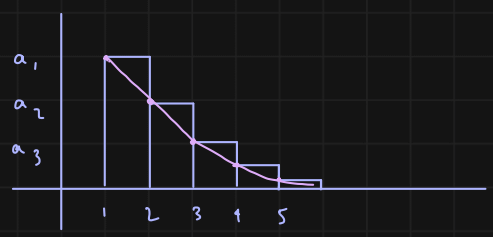
Consider $\sum_{n=1}^{\infty} a_n$

Define a continuous function f such that $f(n) = a_n$ for all $n \in \mathbb{Z}^+$.

We have $\sum_{n=2}^{t} a_n < \int_1^t f(x)\, dx$

Take the limit to infinity of both sides. We get

$$\sum_{n=2}^{\infty} a_n < \lim_{t \to \infty} \int_1^t f(x)\, dx$$



$$\int_1^t f(x)\, dx < \sum_{n=1}^{t} a_n$$

So $\lim_{t \to \infty} \int_1^t f(x)\, dx < \sum_{n=1}^{\infty} a_n$

Theorem — If $f(x)$ is continuous, positive, and decreasing for $x \geq l$, then $\sum_{n=1}^{\infty} f(n)$ converges iff

$\lim_{t \to \infty} \int_1^t f(x)\, dx$ converges

integral test

CONTINUED

Ex Prove the harmonic series diverges.

Let $f(x) = \frac{1}{x}$. $f(x)$ is clearly continuous and positive for $x \geq 1$. Observe that $\frac{1}{n} > \frac{1}{n+1}$, hence it is decreasing. Consider $\lim_{t \to \infty} \int_1^t \frac{1}{x} \, dx$.

$$= \lim_{t \to \infty} \ln|x| \Big|_1^t$$

$$= \lim_{t \to \infty} \ln(t) - \ln(1)$$

$$= \infty$$

So the series diverges by the Integral Test.

Ex Determine if $\sum_{n=2}^{\infty} \frac{e^n}{1+e^{2n}}$ converges or diverges.

Let $f(x) = \frac{e^n}{1+e^{2n}}$. f is continuous and positive for $x \geq 2$. Observe that

$$f'(x) = \frac{e^x(1+e^{2x}) - e(2e^{2x})}{(1+e^{2x})^2}$$

$$= \frac{e(1+e^{2x} - 2e^{2x})}{(1+e^x)^2}$$

$$= \frac{e^x(1-e^{2x})}{(1+e^{2x})^2} < 0 \text{ for } x \geq 2$$

Consider $\lim_{t \to \infty} \int^t \frac{e^x}{1+e^{2x}} \, dx$. Let $u = e^x$.

Then $du = e^x \, dx$. When $x = 2$, $u = e^2$ when $x = t$, $u = e^t$. We get

$$= \lim_{t \to \infty} \int_{e^2}^{e^t} \frac{1}{1+u^2} \, du$$

CONTINUED

$$= \lim_{t \to \infty} \arctan(u) \Big|_{e^2}^{e^t}$$

$$= \lim_{t \to \infty} \left(\arctan(e^t) - \arctan(e^2) \right)$$

$$= \frac{\pi}{2} - \arctan(e^2)$$

So, by the Integral Test, the series converges.

Def'n A series of the form $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is a <u>p - series</u>.

Theorem <u>P - series Test</u>

For a p - series $\sum_{n=1}^{\infty} \frac{1}{n^p}$, if $p \leq 1$ it <u>diverges</u>.

If $p > 1$ it <u>converges</u>.

Ex $\sum_{n=1}^{\infty} \frac{1}{n^{1.00000000000000001}}$ converges by the p - series test since $p = 0.00000000000000001 > 1$.

p - series, p - series test